

VATSAL DHUVAD

Computer Engineering Student | Data Science | Machine Learning | Computer Vision
vatsaldhuvad23@gmail.com | +91 9574788321 | Gandhinagar, Gujarat
Portfolio | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Computer Engineering student specializing in Data Science, Machine Learning, AI/ML, and Computer Vision. Skilled in Python, SQL, data preprocessing, feature engineering, model training, model evaluation, and real-time OpenCV applications. Hands-on project experience includes classification, regression, clustering, and webcam-based face, eye, and smile detection.

TECHNICAL SKILLS

Programming and Databases: Python, SQL, MongoDB, Firebase, Supabase

Data Science Libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn

Machine Learning: Regression, Classification, Clustering, Feature Engineering, Data Preprocessing, Model Training, Model Evaluation

Algorithms and Metrics: Logistic Regression, Linear Regression, Random Forest, Decision Trees, K-Means, Accuracy, Precision, Confusion Matrix, MAE, RMSE, R-squared, Silhouette Score

Computer Vision and Tools: OpenCV, Haar Cascade Classifiers, VS Code, Google Colab, GitHub

EDUCATION

B.E. Computer Engineering, LDRP-ITR, Kadi Sarva Vishwavidyalaya (KSV) - 2023 - Present
Gandhinagar, Gujarat | Currently in 4th Year (7th Semester) | CPI: 6.86

PROJECTS

Student Placement Prediction using Machine Learning - Python, Pandas, NumPy, Scikit-learn, Matplotlib

- Built an end-to-end classification system on a 10,000-row student dataset to predict placement status using Logistic Regression.
- Performed data cleaning, label encoding, feature scaling with StandardScaler, and model evaluation using Accuracy, Precision, and a Confusion Matrix.

House Price Prediction using Machine Learning - Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

- Trained and compared Linear Regression and Random Forest Regressor models to forecast property values using area and location-based input features.
- Cleaned duplicate and missing records, then evaluated results using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared scores.

Customer Segmentation using K-Means Clustering - Python, Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib

- Applied K-Means Clustering with 5 customer groups to identify spending patterns from demographic and behavioral features.
- Used feature selection, scaling, the Elbow Method, and a Silhouette Score of 0.55 to validate cluster quality before dashboard export.

Real-Time Face, Eye, and Smile Detection - Python, OpenCV, Haar Cascade Classifiers, Computer Vision

- Developed a real-time computer vision application using OpenCV and webcam video capture to detect faces, eyes, and smiles from live frames.
- Implemented Haar Cascade classifiers, grayscale conversion, region-of-interest scanning, and labeled bounding boxes for detected facial features.

CERTIFICATIONS

NPTEL / IIT Madras: Python for Data Science - Elite Certification (64%)

Skill India / NSDC: Python Programming; Python for Data Science; Data Science Essentials; AI-ML Engineer

Google Cloud: Gemini for Data Scientists and Analysts; Agent Fundamentals; Engineer AI Agents with Agent Development Kit (ADK)

Anthropic Academy: Building with the Claude API; Claude Code in Action; AI Capabilities and Limitations

HP Foundation: Data Science and Analytics; AI for Beginners; Critical Thinking in the AI Era

Simplilearn: Data Structures and Algorithms in Python; AI Agent for Beginners; Intro to Generative AI Studio